Experiment/Practical 11 Project /Case Study Phase I & Phase II

COURSE PROJECT

PHASE I

Title: Hotel Booking Cancellation Prediction using Machine Learning Algorithms

Aim: To develop and compare machine learning models to predict whether a hotel booking will be canceled based on customer and booking details.

Problem Statement: The objective is to build a classification model that predicts booking cancellation using features such as number of adults, room type, lead time, and price.

This is a binary classification problem:

- 0 → Not Cancelled
- 1 → Cancelled

Literature Work

Sr No	Paper Title	Author	Year	Key Findings
1	Prediction of hotel booking cancellations: integration of ML and probability models	Chen et al.	2023	Combining ML with probabilistic models improves prediction accuracy and interpretability
2	Forecasting Hotel Cancellations using ML	Various (Wiley Journal Study)	2024	Classification models effectively predict cancellations using customer behavior features
3	Hotel booking cancellation prediction using Bayesian models	Jishan M.A., Singh V	2024	Lead time, special requests, and parking strongly influence cancellation probability
4	Detecting short-notice cancellations using ML techniques	C-Sánchez E.	2024	Combining clustering with Random Forest improves prediction performance
5	Hotel booking cancellation and machine learning	Sun J.	2024	Feature engineering and multiple ML models achieve high prediction accuracy

Objectives:

- To predict hotel booking cancellations using machine learning
- To compare performance of multiple classification models
- To identify important factors influencing cancellation

Algorithm:

1. Load dataset
2. Remove irrelevant columns (Booking_ID)
3. Handle missing values
4. Convert date into numerical features
5. Encode categorical variables
6. Split dataset into training and testing
7. Apply feature scaling
8. Perform hyperparameter tuning using GridSearchCV
9. Train models (LR, RF, GB)
10. Evaluate using performance metrics

Expected Outcome:

- Accurate prediction of booking cancellation
- Identification of best-performing model
- Insights into customer booking behavior

PHASE II

Title: Hotel Booking Cancellation Prediction using Machine Learning Algorithms

---

Methodology/ Project Pipeline:

1. Data Collection
2. Data Preprocessing
3. Feature Engineering (date conversion)
4. Encoding categorical variables
5. Train-test split
6. Feature scaling
7. Model training with GridSearchCV
8. Model evaluation
9. Visualization (EDA + ROC curve)

Implementation:

1. **Tools/Libraries Used:**

   - Python
   - Pandas
   - NumPy
   - Scikit-learn
   - Matplotlib
   - Seaborn

2. **Data / Input:**

   The dataset contains hotel booking records with features such as:

   - Number of adults and children
   - Room type
   - Meal type
   - Lead time
   - Market segment
   - Average price
   - Special requests

   Target variable:

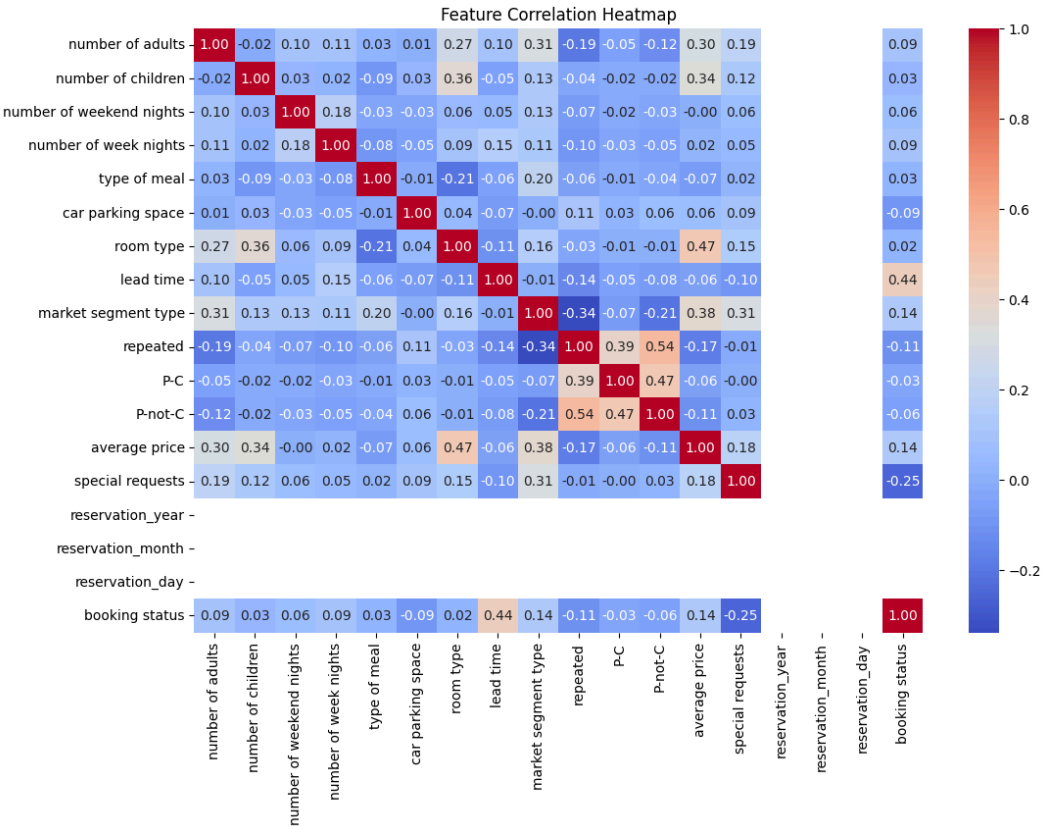
Booking Status (0 = Not Cancelled, 1 = Cancelled)

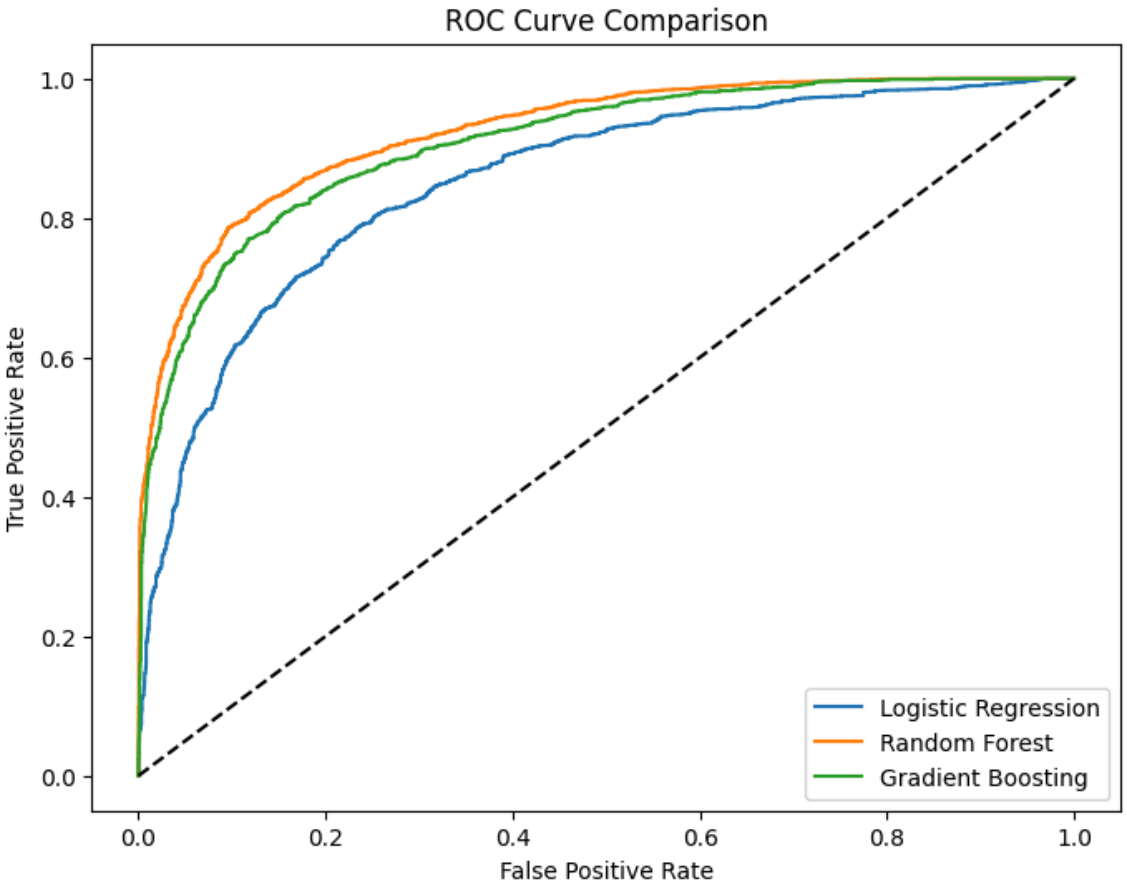
3. Results:

Logistic Regression: 80.33%

Random Forest: 86.17%

Gradient Boosting: 84.82%


Feature Correlation Heatmap
number of adults
number of children
number of weekend nights
number of week nights
type of meal
car parking space
room type
lead time
market segment type
repeated
P-C
P-not-C
average price
special requests
reservation_year
reservation_month
reservation_day
booking status
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


ROC Curve Comparison
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression
Random Forest
Gradient Boosting

Key Observations:

- Random Forest achieved the highest accuracy (86.17%)
- Logistic Regression provided baseline performance
- Gradient Boosting performed slightly lower than Random Forest
- Model performance improved due to preprocessing and tuning

Insights from Data

- Higher lead time increases cancellation probability
- Customers with more special requests are less likely to cancel
- Pricing influences cancellation behavior
- Certain room types and segments show higher cancellation rates

---

Conclusion: The project successfully implemented machine learning models to predict hotel booking cancellations. After preprocessing and hyperparameter tuning, Random Forest emerged as the best-performing model with the highest accuracy.

The study highlights the importance of features such as lead time, pricing, and customer behavior in predicting cancellations. The developed model can help hotels optimize booking strategies and reduce revenue loss.

Future Scope:

- Use deep learning models for better performance
- Deploy model as a web application
- Integrate real-time booking data
- Apply advanced feature selection techniques
- Improve model using larger datasets